MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 4 Solution

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1. Show the limit of $a_n = \sqrt[n]{n}$ by definition.

   Solution. Let $\sqrt[n]{n} = 1 + d(n)$, where $d(n) > 0$, and

   $$n = (1+d)^n = 1 + nd + \frac{n(n-1)}{2}d^2 + \cdots + d^n > \frac{n(n-1)}{2}d^2 \implies d < \sqrt{\frac{2}{n-1}} < \varepsilon$$

   ■

2. Find $\lim_{n\to+\infty} \left(\frac{n+1}{n-1}\right)^n$.

   Solution.

   $$\lim_{n\to+\infty} \left(\frac{n+1}{n-1}\right)^n = \lim_{n\to+\infty} \frac{(1+\frac{1}{n})^n}{(1+\frac{-1}{n})^n} = \frac{e}{e^{-1}} = e^2$$

   ■

3. Find a sequence with two convergent subsequences of distinct limits, then find one with infinitely many distinct subsequence limits.

   Solution.
   $a_n = (-1)^n$, let $f(n) : \mathbb{N} \to \mathbb{N}\times\mathbb{N}$ be a bijection. Then all numbers in the set $\{\frac{p}{q}, \frac{10p}{10q}, \frac{100p}{100q}, \frac{1000p}{1000q} \cdots\}$ will belong to a subsequence n_{n_k} of $b_n = f(n)$ which $n_{n_k} \to \frac{p}{q}$ for any $\frac{p}{q} \in \mathbb{Q}$. One can show that b_n also has subsequence converges to any irrational numbers. ■

4. Let (x_n) and (y_n) be bounded sequences. Prove that there exists a common subsequence (n_k) such that both (x_{n_k}) and (y_{n_k}) converge.

   Solution.
   By Bolzano Weierstrass theorem, $x_{n_k} \to L_x$. y_{n_k} is bounded, by BWT again, $y_{n_{k_h}} \to L_y$. We also have $x_{n_{k_h}} \to L_x$. ■

5. Find if the following series converge.

   (a) $\displaystyle\sum_1^\infty \frac{n}{n+2-\cos^2 n}$

   (b) $\displaystyle\sum_1^\infty \frac{e^{-n}}{n+\cos^2 n}$

   Solution.

   (a) $a_n \to 1$, divergent by term test.

   (b) Converge, by limit comparison with $\frac{1}{n^2}$,

$$\frac{\frac{e^{-n}}{n+\cos^2 n}}{n^{-2}} = \frac{n^2 e^{-n}}{n+\cos^2 n} \to 0$$

■

6. Let $\sum a_n = +\infty$ be a positive divergent series, i.e. $a_n > 0$, and $\sum b_n = \sum (-1)^n a_n = L < \infty$ is convergent. Explain if there is an rearrangement f of b_n such that $\sum b_{f(n)} = 2241$.

   Solution.

   https://en.wikipedia.org/wiki/Riemann_series_theorem ■

— *End* —